IIT ROPAR · AI101 MINI-PROJECT · GROUP 47

Predicting Vehicle CO_2 Emissions

using Multi-Variable Linear Regression

Our model looks at a car's fuel use, weight, power, fuel type and gearbox, and predicts how many grams of CO_2 it releases for every kilometre it drives.

GitHub repo: https://github.com/prit-iitrpr/AI101-Project.git

0.96	4.8 g/km
R^2 on test data	average error (MAE)
7,385	**α = 0.3**
vehicle records	best learning rate

OUR TEAM (GROUP 47)

EXECUTION LEAD
Vora Prit A.
2026AIB1097
Data cleaning & model code

VIDEO LEAD
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2026EEB1351
Demo recording & voiceover

DOCUMENTATION LEAD
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Maths proofs & documentation

PPT LEAD
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Slide design & presentation

The problem and the data

WHAT & WHY

Cars release CO_2 from the exhaust. Transport causes about 24% of direct global CO_2. If we can predict CO_2 from a car's specs, regulators can check cars without a full lab test.

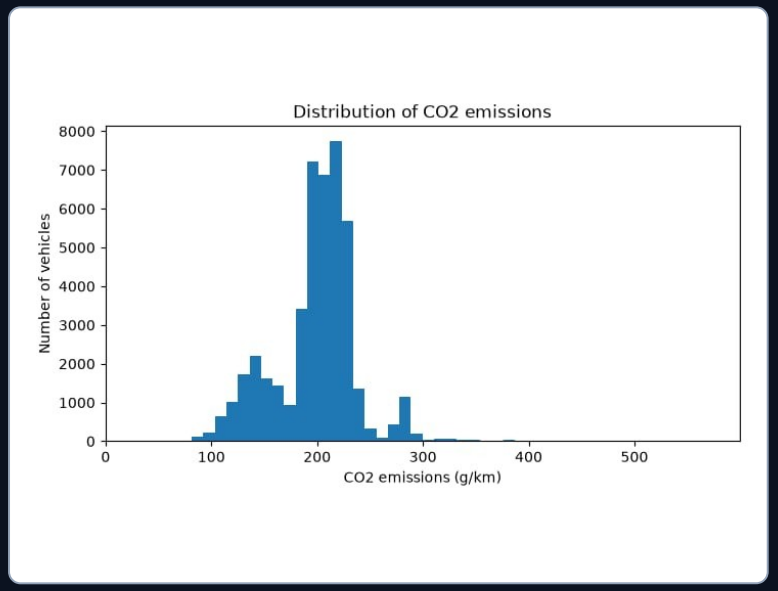
DATASET SOURCE

Public vehicle emissions (car-labelling) data. Every row is one car model with its fuel use, mass, power, fuel type, gearbox and body type. Link is in our README.

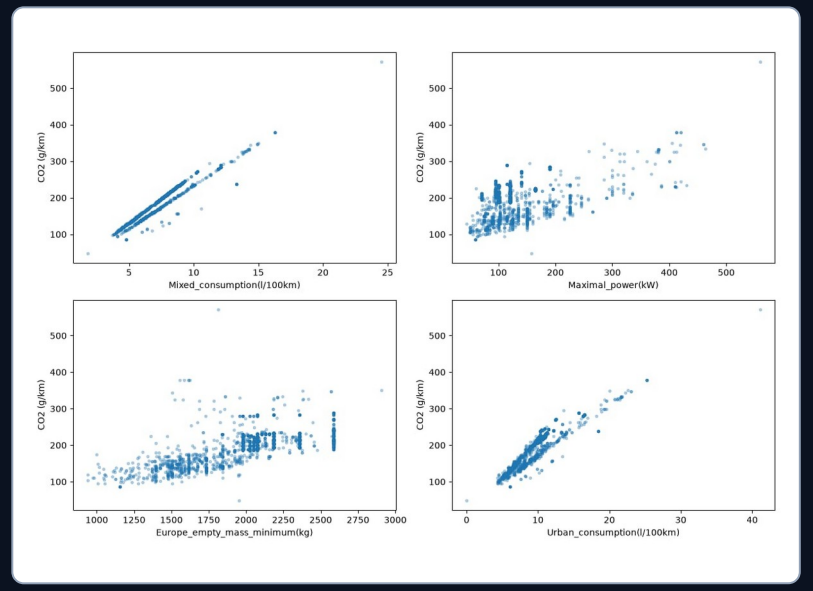
SIZE & COLUMNS

7,385 vehicle records. Output we predict: CO_2 in g/km. Inputs: fuel consumption, mass, power, gears, fuel type, hybrid yes/no, gearbox type and body type.

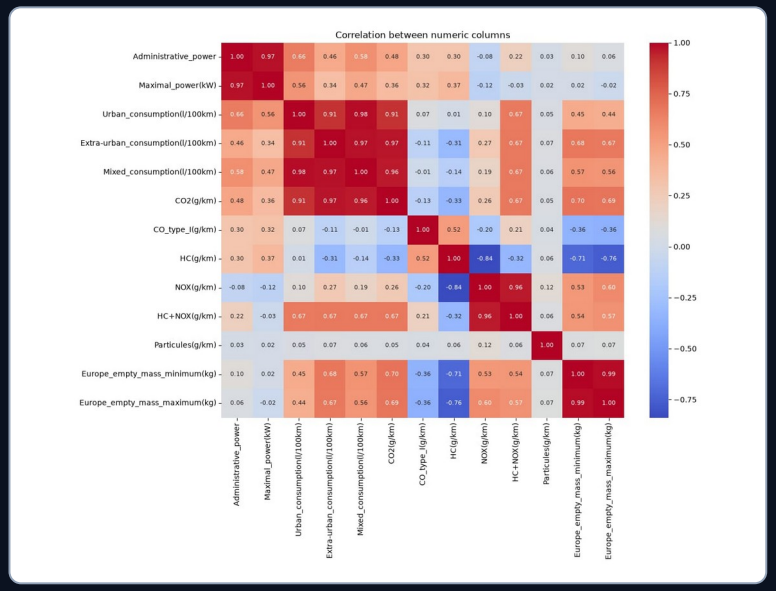
WHAT DOES THE DATA LOOK LIKE?



Most cars emit 180–230 g/km (average ≈ 200). A smaller group of efficient cars sits near 140.



Fuel use and CO_2 rise together in a near-straight line, so a linear model is a good fit.



Dark red = strong link. Fuel use vs CO_2 is 0.96, the strongest of all.

Our approach: data → model → prediction

1 — Clean
Remove duplicates, fill missing values, drop extreme outliers.

2 — One-hot encode
Turn fuel type, gearbox and body type into 0/1 columns.

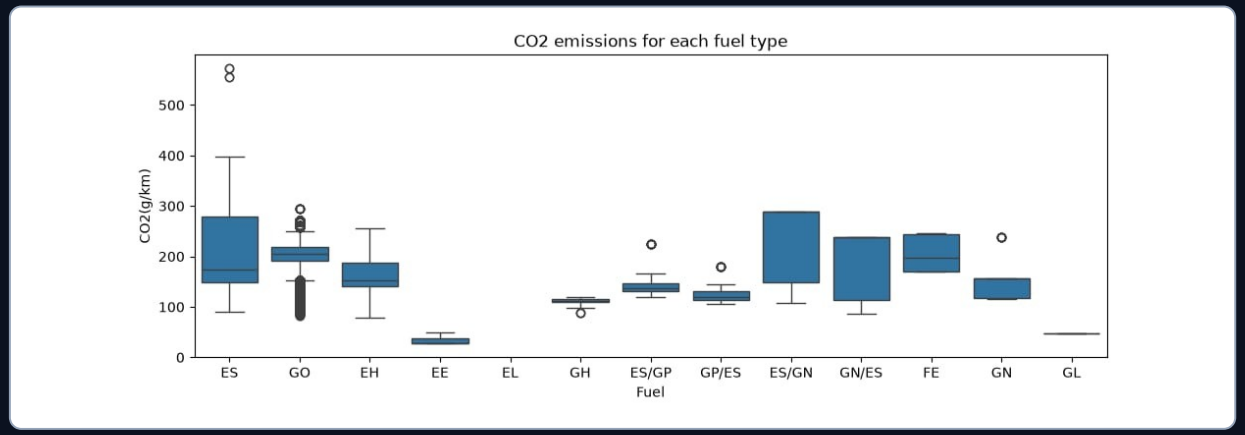
3 — Z-score scale
Put every number on the same scale (mean 0, spread 1).

4 — Train
Gradient descent adjusts weights θ to shrink the error.

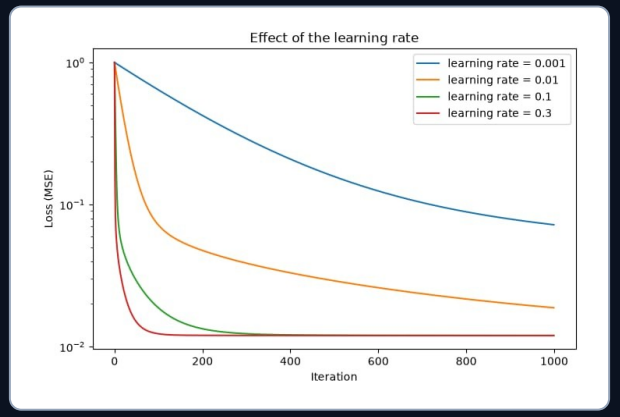
5 — Predict
$\hat{y} = \theta^T x$ gives the CO_2 of cars the model has never seen.

$$h(x) = \theta_0 + \theta_1 x_1 + \ldots + \theta_n x_n = \theta^T x$$
$$J(\theta) = (1/2m) \Sigma (h(x) - y)^2 \quad \text{(MSE loss)}$$
$$\theta_j := \theta_j - \alpha \cdot (1/m) \Sigma (h(x) - y) \cdot x_j$$

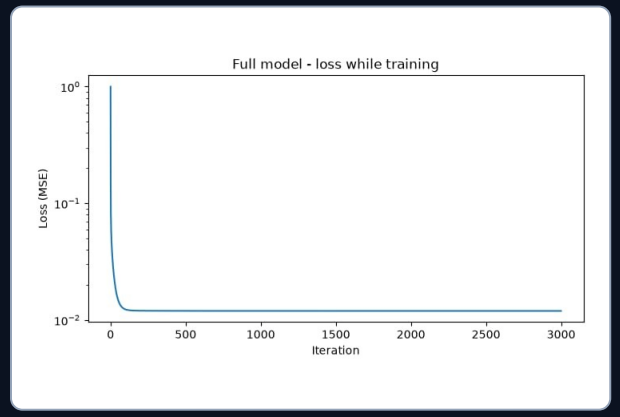
Python 3.10 · NumPy · Pandas · Scikit-Learn · Matplotlib



Fuel type matters: diesel (GO) is steady, petrol (ES) varies a lot, electric (EE) is near zero. That is why we one-hot encode it.



Learning rate: α = 0.3 reaches the lowest error fastest.



Loss flattens after ~150 steps: the model has learned.

Results: how good is the model?

R^2 (TEST DATA)

0.96

of CO_2 differences explained

MEAN ABSOLUTE ERROR

4.8 g/km

average miss per car

ROOT MEAN SQUARED ERROR

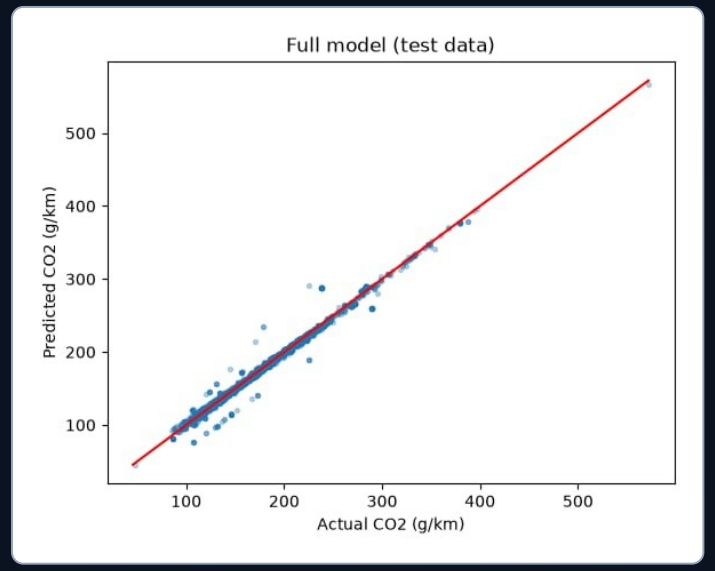
6.2 g/km

big misses count extra

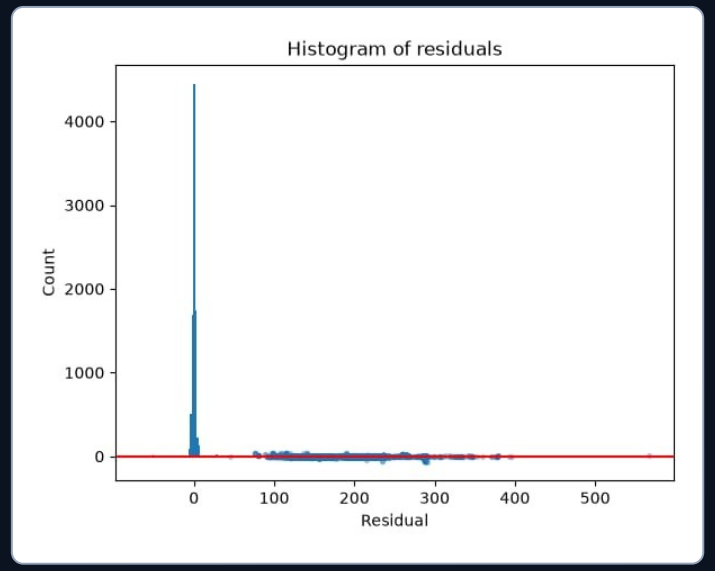
WITHOUT FUEL USE

R^2 0.81

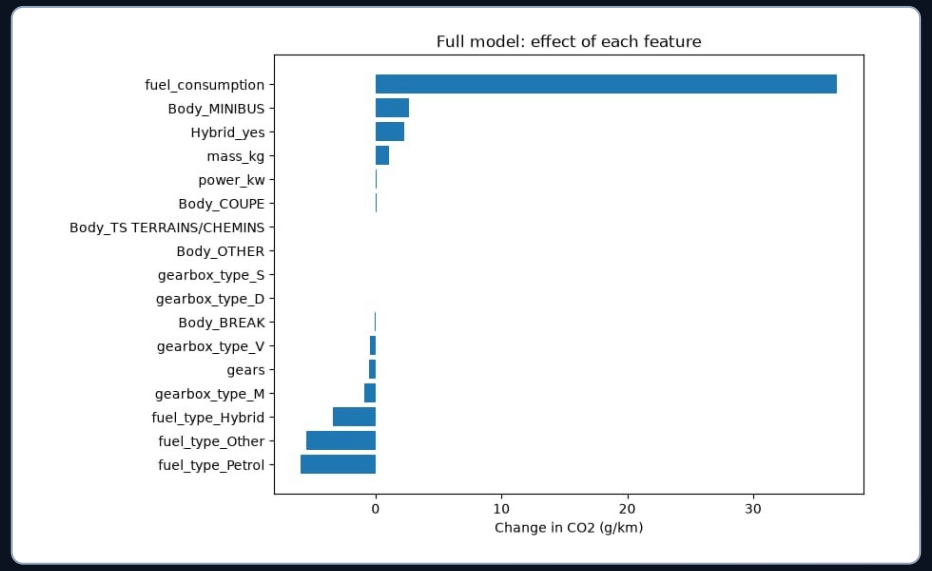
MAE 14.2 g/km (3× worse)



Dots sit right on the red line: predicted CO_2 matches real CO_2 almost perfectly.



Most errors are right at 0. Very few cars are badly off.



Fuel use is by far the biggest driver (about +37 g/km). Fuel type, hybrid and body type only nudge the answer.

Takeaway: fuel consumption explains most of the CO_2; the other features fine-tune the last few grams.

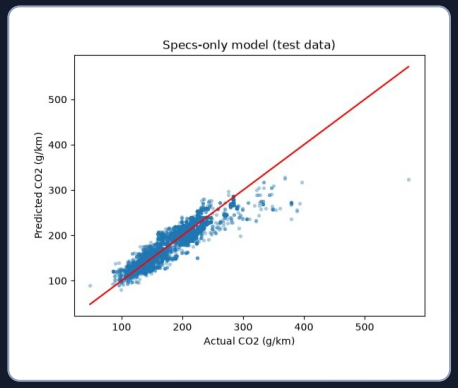
What we learned and who did what

WHAT WORKED

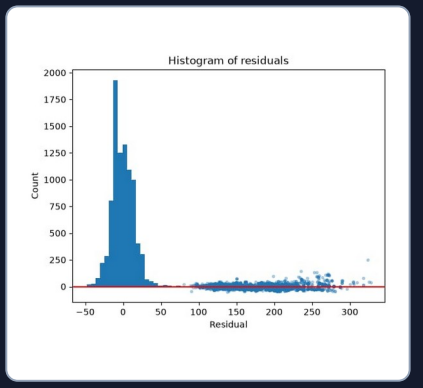
- One-hot encoding handled fuel, gearbox and body type with no extra tricks.
- Z-score scaling + α = 0.3 made training fast and stable.
- Full model reached $R^2 = 0.96$.

WHAT DIDN'T WORK

- Specs-only model (no fuel use): R^2 falls to 0.81, error triples.
- Sports cars and heavy minivans had bigger errors (unusual engines).



Specs-only: more scatter



Specs-only: wider errors

WHO DID WHAT

Vora Prit A.
`2026AIB1097`
Data cleaning, outlier treatment, batch gradient descent code

Geetansh Verma
`2026EEB1351`
Video demo script, screen recording, voiceover, editing

Shetye Ishaan S.
`2026MCB1568`
MSE loss derivation, LaTeX README and GitHub docs

Ashish Meena
`2026ESB1473`
Slide design, data presentation, project defence

AI-use disclosure: Generative AI was used for debugging help, chart styling and slide formatting.